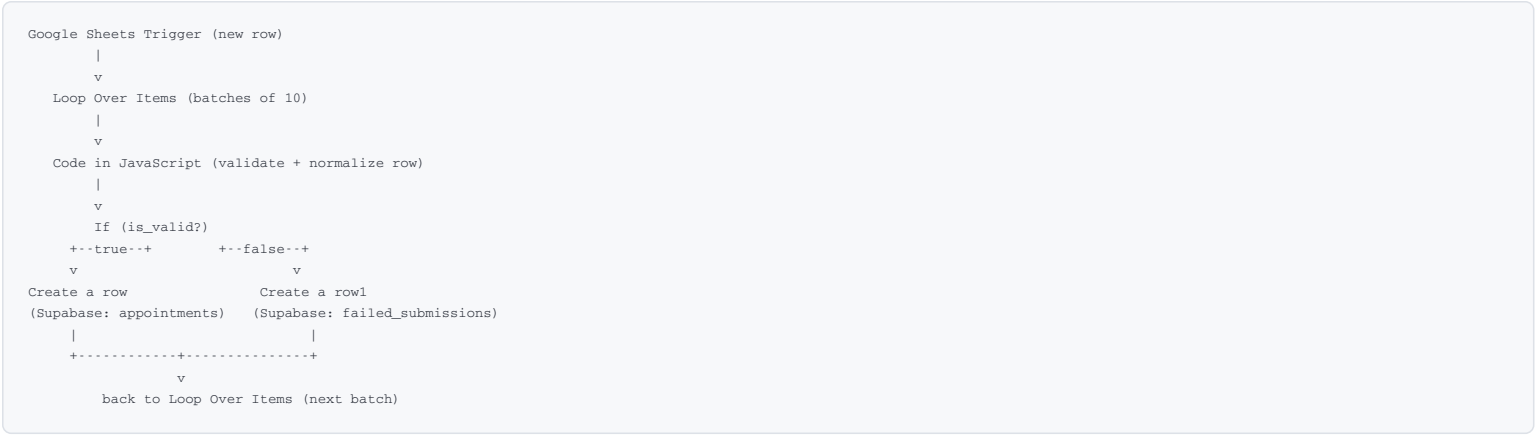


📅 Appointments Automation Workflow (n8n)

Google Sheets → validation → Supabase, with automatic error logging

This n8n workflow automates appointment booking intake from a **Google Form → Google Sheet**, validates and cleans the submitted data, and stores valid appointments in **Supabase**. Invalid submissions are logged separately for review instead of being dropped silently.

🔄 Workflow Overview



What it does:

- Watches a Google Sheet (linked to a Google Form) for new appointment booking rows.
- Processes rows in batches of 10 using **Loop Over Items**.
- Runs each row through a **Code** node that:
 - Normalizes header keys (handles spacing/casing mismatches).
 - Validates full name (no digits, only letters/spaces/ `.` / `'` / `-`).
 - Cleans and validates the Indian mobile number (10 digits, strips +91/0 prefixes, must start with 6-9).
 - Validates gender and doctor fields are present.
 - Validates the appointment date is a real date and not in the past.
 - Validates the appointment time matches `HH:MM:SS` format.
 - Generates a `row_hash` (simple hash of name + phone + day + time) for de-duplication.
 - Returns a structured object with `is_valid` and a semicolon-separated `errors` string.
- An **If** node branches on `is_valid` :
 - True** → inserts a clean row into the Supabase `appointments` table.
 - False** → inserts the errors + raw row JSON into the Supabase `failed_submissions` table.
- Loops back until all batched rows are processed.

🔧 Part 1 — n8n Installation

You can run n8n via **npm**, **Docker**, or **n8n Cloud**. Self-hosted via Docker is recommended for production.

Option A: Docker (recommended)

```
docker volume create n8n_data

docker run -it --rm \
  --name n8n \
  -p 5678:5678 \
  -e GENERIC_TIMEZONE="Asia/Kolkata" \
  -e TZ="Asia/Kolkata" \
  -v n8n_data:/home/node/.n8n \
  docker.n8n.io/n8nio/n8n
```

Then open `http://localhost:5678` and complete the first-time owner account setup.

Option B: npm (quick local test)

```
npm install -g n8n
n8n start
```

Option C: docker-compose (persistent, easier to manage)

```
version: "3.7"
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    restart: always
    ports:
      - "5678:5678"
    environment:
      - GENERIC_TIMEZONE=Asia/Kolkata
      - TZ=Asia/Kolkata
      - N8N_SECURE_COOKIE=false
    volumes:
      - n8n_data:/home/node/.n8n

volumes:
  n8n_data:
```

```
docker compose up -d
```

Importing this workflow

1. Open n8n → **Workflows** → **Add workflow** → ⓘ menu → **Import from File**.
2. Select `Appointments.json` from this repo.
3. Reconnect the credentials (see below) since credential IDs do not transfer between instances.

📖 Part 2 — Supabase Configuration

2.1 Create a Supabase project

1. Go to **supabase.com** → **New Project**.
2. Note down your **Project URL** and **service_role** / **anon API key** (Project Settings → API).

2.2 Create the required tables

Run this in the Supabase **SQL Editor**:

```
-- Table for successfully validated appointments
create table public.appointments (
  id uuid primary key default gen_random_uuid(),
  full_name text not null,
  gender text,
  personnel_number text not null,
  doctor text,
  appointment_day date,
  appointment_time time,
  sheet_row_hash text unique,
  created_at timestampz default now()
);

-- Table for rejected / invalid submissions
create table public.failed_submissions (
  id uuid primary key default gen_random_uuid(),
  errors text,
  raw_data jsonb,
  created_at timestampz default now()
);
```

🔗 `sheet_row_hash` has a **unique** constraint to help prevent duplicate inserts if the same row is processed twice. If you want the workflow to silently skip duplicates instead of erroring, add an `upsert / on conflict do nothing` step, or handle it in the Code node.

2.3 (Optional) Row Level Security

If RLS is enabled on your project by default, either:

- Disable RLS for these two tables (simplest, fine for a backend-only automation), **or**
- Add a policy allowing inserts from the `service_role` key (which bypasses RLS by default anyway).

```
alter table public.appointments disable row level security;
alter table public.failed_submissions disable row level security;
```

🔑 Part 3 — Credentials Configuration in n8n

This workflow needs **two credentials**:

3.1 Google Sheets Trigger (OAuth2)

Used by: **Google Sheets Trigger** node.

1. In n8n: **Credentials** → **New** → **Google Sheets Trigger OAuth2 API**.
2. Follow n8n's guide to create a Google Cloud OAuth Client (Google Cloud Console → APIs & Services → Credentials → OAuth 2.0 Client ID, type "Web application").
3. Enable the **Google Sheets API** in your Google Cloud project.

- 4. Add the n8n redirect URI shown in the credential screen to your OAuth Client's **Authorized redirect URIs**.
- 5. Paste the **Client ID** and **Client Secret** into the n8n credential, then click **Connect my account** and authorize.
- 6. In the **Google Sheets Trigger** node, select:
 - **Document**: your Google Form's linked response sheet.
 - **Sheet**: `Form responses 1`.
 - **Event**: `Row Added`.
 - **Poll time**: `Every Minute` (adjust as needed).

3.2 Supabase API

Used by: **Create a row** and **Create a row1** nodes.

- 1. In n8n: **Credentials** → **New** → **Supabase API**.
- 2. **Host**: your Supabase Project URL (e.g. `https://xxxxx.supabase.co`).
- 3. **Service Role Secret** (or anon key, if RLS allows inserts): from Supabase → **Project Settings** → **API**.
- 4. Save, then attach this credential to both Supabase nodes.

⚠ Treat the **service_role** key as a secret — never commit it to git. Use n8n's credential store (already excluded from the exported workflow JSON) or environment variables if self-managing.

🔧 Part 4 — Node-by-Node Build Steps

If you're building this workflow from scratch instead of importing the JSON:

1. Google Sheets Trigger

- Add node **Google Sheets Trigger**.
- Set **Event** = `Row Added`.
- Set **Poll Times** = Every Minute.
- Select the **Document** = your Google Form responses spreadsheet.
- Select the **Sheet** = `Form responses 1`.
- Attach the Google Sheets Trigger OAuth2 credential.

2. Loop Over Items (Split In Batches)

- Add node **Loop Over Items** (`n8n-nodes-base.splitInBatches`).
- Set **Batch Size** = `10`.
- Leave **Reset** = off.
- Connect **Google Sheets Trigger** → **Loop Over Items**.
- This node has two outputs: `done` (index 0) and `loop` (index 1). Connect the **loop** output forward to the Code node — the **done** output is left unconnected (used only when all batches finish).

3. Code in JavaScript

- Add a **Code** node, mode = **Run Once for Each Item**.
- Connect **Loop Over Items (loop output)** → **Code in JavaScript**.
- Paste the validation script (see `Apointments.json` for the exact code), which:
 - Normalizes the incoming Google Form headers via a case/whitespace-insensitive `getField()` helper.
 - Validates: name, phone (Indian 10-digit mobile), gender, doctor, appointment date (not in the past), appointment time format (`HH:MM:SS`).
 - Builds an ISO `appointment_day` (`YYYY-MM-DD`).
 - Computes a `row_hash` for de-duplication.
 - Outputs `{ full_name, gender, personnel_number, doctor, appointment_day, appointment_time, row_hash, is_valid, errors }`.

4. If

- Add an **If** node.
- Condition: `{{ $json.is_valid }}` **equals** `true` (string comparison, loose type validation enabled).
- Connect **Code in JavaScript** → **If**.

5. Create a row (Supabase — appointments)

- Add a **Supabase** node named **Create a row**.
- **Table**: `appointments`.
- Map fields:

Supabase field	Value
full_name	<code>{{ \$json.full_name }}</code>
gender	<code>{{ \$json.gender }}</code>
personnel_number	<code>{{ \$json.personnel_number }}</code>
doctor	<code>{{ \$json.doctor }}</code>
appointment_day	<code>{{ \$json.appointment_day }}</code>
appointment_time	<code>{{ \$json.appointment_time }}</code>
sheet_row_hash	<code>{{ \$json.row_hash }}</code>

- Connect **If (true output) → Create a row**.
- Attach the Supabase API credential.

6. Create a row1 (Supabase — failed_submissions)

- Add a **Supabase** node named **Create a row1**.
- **Table:** `failed_submissions`.
- Map fields:

Supabase field	Value
errors	<code>{{ \$json.errors }}</code>
raw_data	<code>{{ JSON.stringify(\$json) }}</code>

- Connect **If (false output) → Create a row1**.
- Attach the same Supabase API credential.

7. Close the loop

- Connect **Create a row → Loop Over Items**.
- Connect **Create a row1 → Loop Over Items**.
- This sends control back to **Loop Over Items** so it processes the next batch of 10 rows until the sheet's new rows are exhausted.

8. Activate the workflow

- Toggle the workflow **Active** in the top-right corner so the Google Sheets Trigger starts polling every minute.

✓ Testing Checklist

- ☐ Submit a test entry via the Google Form.
- ☐ Confirm a new row appears in `Form responses 1`.
- ☐ Confirm the workflow execution runs and completes without errors.
- ☐ Confirm a valid submission lands in the Supabase `appointments` table.
- ☐ Submit an intentionally invalid entry (e.g. bad phone number) and confirm it lands in `failed_submissions` with a populated `errors` field.

📁 Repository Contents

File	Description
Apointments.json	Exported n8n workflow (import via n8n UI)
README.md	This documentation

🔒 Security Notes

- Do not commit real Supabase service keys or Google OAuth client secrets to git — n8n's exported JSON only stores credential **references** (IDs/names), not secret values, so this is safe by default. Just make sure you don't paste actual keys anywhere else in the repo (e.g. `.env` files).
- If you fork this repo, you must recreate/reconnect both credentials in your own n8n instance — they will not be pre-filled.